

Research and Development Individual Worklog

This document provides a detailed log of activities, grouped by project week.

S1 - Week 1 (March 3 - March 9, 2025)

- **Workshop:** Week 1 - Workshop
 - **Type:** Workshop
 - **Date:** March 6, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Week 1 - Workshop Notes.docx](#)
 - **Total Hours for S1 - Week 1:** 2h 0m
-

S1 - Week 2 (March 10 - March 16, 2025)

- **Offsite:** Working out timetables
 - **Type:** Offsite
 - **Date:** March 10, 2025
 - **Time:** 12:00 PM - 1:00 PM
 - **Duration:** 1h 0m
 - [Timetables.xlsx](#)
 - [Timetables](#) (Folder with screenshots of our timetables)
- **Offsite:** Stakeholder Register
 - **Type:** Offsite
 - **Date:** March 11, 2025
 - **Time:** 12:00 PM - 12:30 PM
 - **Duration:** 0h 30m
 - [02. Stakeholder Register.docx](#)
- **Offsite:** Worked on paperwork / Met the team
 - **Type:** Offsite
 - **Date:** March 13, 2025
 - **Time:** 12:30 PM - 2:00 PM
 - **Duration:** 1h 30m
- **Meeting:** Client / Mentor Meeting - Cancelled
 - **Type:** Meeting
 - **Date:** March 13, 2025
 - **Time:** 2:00 PM - (Cancelled)
 - **Duration:** 0h 0m
 - Due to the meeting clashing with the lecture it was cancelled

- **Workshop:** Week 2 - Workshop
 - **Type:** Workshop
 - **Date:** March 13, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Week 2 - Workshop Notes.docx](#)
 - **Offsite:** Filling out paperwork
 - **Type:** Offsite
 - **Date:** March 15, 2025
 - **Time:** 11:00 AM - 12:40 PM
 - **Duration:** 1h 40m
 - **Total Hours for S1 - Week 2:** 6h 40m
-

S1 - Week 3 (March 17 - March 23, 2025)

- **Meeting:** Online Meeting
 - **Type:** Meeting
 - **Date:** March 17, 2025
 - **Time:** 1:30 PM - 3:00 PM
 - **Duration:** 1h 30m
 - Unofficial meeting with only some of the members to try to get to know each other, but most couldn't make it.
- **Meeting:** Online Meeting - Kick off
 - **Type:** Meeting
 - **Date:** March 18, 2025
 - **Time:** 7:00 PM - 8:30 PM
 - **Duration:** 1h 30m
 - [Meeting Minutes 18-03-2025 \(Team Kick-off\).docx](#)
- **Offsite:** Filling out assigned paperwork
 - **Type:** Offsite
 - **Date:** March 19, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - Was assigned to work on the stakeholder register and management plan.
 - [02. Stakeholder Register.docx](#)
 - [04. Stakeholder Management Plan.docx](#)

- **Meeting:** Meeting - Client / Mentor

- **Type:** Meeting
- **Date:** March 20, 2025
- **Time:** 10:00 AM - 11:00 AM
- **Duration:** 1h 0m
 - [Meeting Minutes 20-03-2025 \(Client Kick-off\).docx](#)
 - New member to team, Kylie. Discussed setting up the project's physical environment, which includes four computers two configured as routers and three networks for sender-to-receiver communication, utilizing Fedora, Ubuntu, and Kali Linux. Testing will begin with IPv4 (TCP and UDP) Win, Zafar, and Thomas, followed by IPv6 testing by Larissa, Kylie, Nathan, and Charmi, with all results, including a log file and Excel findings, to be submitted. Major part to the project is establishing a project methodology first, requiring a comparison of three approaches, justification for the chosen one (possibly drawing from SDLC), and realistic cost estimations with references. Regular mentor/client meetings are scheduled—weekly until the Project Proposal submission on April 4th (with a 10-page limit and subsequent presentation), then fortnightly. The AUT IT tech team to be added to Teams for hardware specification and equipment access.

- **Meeting:** Meeting - After Client / Mentor Meeting

- **Type:** Meeting
- **Date:** March 20, 2025
- **Time:** 11:00 AM - 2:00 PM
- **Duration:** 3h 0m
 - [Meeting Minutes 20-03-2025 \(Team\).docx](#)
 - The team was considering a hybrid approach for the project methodology, specifically combining Agile with SDLC principles. Planned to create a comparison table outlining the advantages and challenges of three different methodologies before selecting the ideal one and defining its phases. The Project Proposal is established as the first milestone, with a target completion date of March 31st to allow ample time for review, editing, and compilation before its April 4th deadline. To ensure active participation, each task will be assigned to at least two team members. The team intends to work on the presentation PowerPoint slides once the proposal is complete.

- **Workshop:** Week 3 - Workshop

- **Type:** Workshop
- **Date:** March 20, 2025
- **Time:** 2:00 PM - 4:00 PM
- **Duration:** 2h 0m
 - [Week 3 - Workshop Notes.docx](#)

- **Total Hours for S1 - Week 3:** 11h 0m

S1 - Week 4 (March 24 - March 30, 2025)

- **Offsite:** Updating Proposal Paperwork
 - **Type:** Offsite
 - **Date:** March 25, 2025
 - **Time:** 12:00 PM - 2:00 PM
 - **Duration:** 2h 0m
 - Updated Stakeholder Register
 - Researched a bit around what D-ITG is, and how to install on linux.
 - Added more information to the proposal document
- **Offsite:** More Proposal paperwork
 - **Type:** Offsite
 - **Date:** March 26, 2025
 - **Time:** 12:00 PM - 2:00 PM
 - **Duration:** 2h 0m
 - Added more to the proposal document
- **Offsite:** Converting Logbook into a word doc
 - **Type:** Offsite
 - **Date:** March 27, 2025
 - **Time:** 11:00 AM - 1:00 PM
 - **Duration:** 2h 0m
- **Meeting:** Mentor meeting
 - **Type:** Meeting
 - **Date:** March 27, 2025
 - **Time:** 1:00 PM - 2:00 PM
 - **Duration:** 1h 0m
 - [Meeting Minutes 27-03-2025 \(Supervisor\).docx](#)
 - The team is busy refining the Project Proposal, which needs a version control table and Table of Contents, with all sections requiring concise paragraphs and consistent formatting. The Terms of Reference will outline the problem, purpose, and opportunity, while the Objectives and Scope will copy objectives directly from the brief, detail functional and non-functional requirements, and clarify deliverables, with further scope details and out-of-scope items relegated to an appendix. A Skills Analysis matrix (summarized in the main document with a full version in the appendix) and an upskilling plan are also mandatory. The team must prepare the Client Contract for signature, identify roles based on their chosen Project Management Methodology (which all members must understand), present the Team Schedule in a table, and compare IPERF and DITG. The Cost Breakdown in the proposal will be a summary table with detailed references in the appendices. Daniel has suggested completing the first draft by Sunday for review and feedback, and he'll schedule a meeting for Thursday, April 3rd, 2025, at 1 PM, for a proposal review and mock presentation. Finally, the team needs to contact the IT technician about required hardware and equipment for performance testing.

- **Workshop:** Week 4 - Workshop
 - **Type:** Workshop
 - **Date:** March 27, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Week 4 - Workshop Notes.docx](#)
 - **Offsite:** Working on the proposal
 - **Type:** Offsite
 - **Date:** March 28, 2025
 - **Time:** 11:00 AM - 2:00 PM
 - **Duration:** 3h 0m
 - Created a network typology image showing how the typology will work and explaining it.
 - **Meeting:** Online Meeting
 - **Type:** Meeting
 - **Date:** March 28, 2025
 - **Time:** 6:00 PM - 8:00 PM
 - **Duration:** 2h 0m
 - [WBS 2.docx](#)
 - The team got in an online call to complete more of the proposal together as we can ask questions easier and get answers faster. Once of the focuses was working on the WBS and trying to complete it as then we could work on the Gantt Chart.
 - **Offsite:** More work on proposal
 - **Type:** Offsite
 - **Date:** March 29, 2025
 - **Time:** 11:00 AM - 6:00 PM
 - **Duration:** 7h 0m
 - Worked on adding additional information to the proposal document
 - **Offsite:** Worked on Gantt Chart
 - **Type:** Offsite
 - **Date:** March 30, 2025
 - **Time:** 12:00 AM - 4:00 AM
 - **Duration:** 4h 0m
 - Worked on the scheduling of the project.
 - [09. Gantt Chart.mpp](#)
 - **Total Hours for S1 - Week 4:** 25h 0m
-

S1 - Week 5 (March 31 - April 6, 2025)

- **Offsite:** Worked on Proposal
 - **Type:** Offsite
 - **Date:** March 30, 2025
 - **Time:** 12:00 PM - 3:00 PM
 - **Duration:** 3h 0m
 - Worked on adding additional information to the proposal document
- **Offsite:** Work on Proposal
 - **Type:** Offsite
 - **Date:** March 31, 2025
 - **Time:** 12:00 PM - 3:00 PM
 - **Duration:** 3h 0m
 - Worked on adding additional information to the proposal document
- **Meeting:** Online Team Meeting
 - **Type:** Meeting
 - **Date:** March 31, 2025
 - **Time:** 4:00 PM - 6:00 PM
 - **Duration:** 2h 0m
 - [Meeting Minutes 31-03-2025 \(Team\).docx](#)
 - For the upcoming presentation, Charmi and Kylie were tasked to create the PowerPoint slideshow, with a team practice session planned before Thursday's mock-up. The team has finalized a hybrid Waterfall-Scrum approach for the project methodology, utilizing Waterfall for the first two phases and Scrum for the remaining two phases, each broken into three sprints corresponding to a Linux OS; there's ongoing discussion about whether STLC is a methodology or a framework, given the client's initial emphasis on SDLC. The Project Proposal document requires careful version control (including comments on changes), a Milestone Report in the Appendix, and a detailed Team Schedule; the Terms of Reference section specifically needs to be a single paragraph with headers removed, and the team should prioritize using "Evaluation" over "Testing" throughout the document. All meeting minutes and agendas will be appended to the Team Schedule, and the Labour Breakdown Table will be summarized in a short paragraph within the proposal, with full details in the Appendix, ensuring all APA 7th references are alphabetized. The proposal document is slated for submission on Friday, following a final team review and feedback incorporation from the supervisor on Thursday, with the Team Leader handling the submission. Finally, the Client Contract and Project Charter need to be signed off, and the team is organizing their portfolio's Gantt Chart files, ideally consolidating them into one.
- **Offsite:** Took notes for Week 5 Workshop from slides
 - **Type:** Offsite
 - **Date:** April 3, 2025
 - **Time:** 2:00 PM - 3:00 PM
 - **Duration:** 1h 0m
 - [Week 5 - Workshop Notes.docx](#)
- **Sick Week:** From 1st to 4th April
 - **Duration:** 0h 0m
 - Was not feeling well for the week, so stayed home to recover.
- **Total Hours for S1 - Week 5:** 9h 0m

S1 - Week 6 (April 7 - April 13, 2025)

- **Meeting:** Final Finish ups of Proposal
 - **Type:** Meeting
 - **Date:** April 6, 2025
 - **Time:** 10:00 PM - April 7, 2025 12:00 AM
 - **Duration:** 2h 0m
 - The team joined an online call and worked on the finalizing of the proposal till the deadline to make sure that it was going to be completed to the best of our ability.
 - **Meeting:** Online Presentation - Practice
 - **Type:** Offsite
 - **Date:** April 7, 2025
 - **Time:** 7:00 PM - 8:30 PM
 - **Duration:** 1h 30m
 - All practiced what we were going to say in the presentation in an online call so that we were prepared for the day.
 - **Meeting:** Pre-presentation meeting
 - **Type:** Meeting
 - **Date:** April 10, 2025
 - **Time:** 9:30 AM - 11:00 AM
 - **Duration:** 1h 30m
 - All practiced what we were going to say in the presentation in person so that we were prepared for what to say in the presentation.
 - **Meeting:** Proposal Presentation
 - **Type:** Meeting
 - **Date:** April 10, 2025
 - **Time:** 11:00 AM - 12:00 PM
 - **Duration:** 1h 0m
 - Official presentation of our proposal in front of Supervisor, Client, and Marker.
 - **R&D Lab:** Trying to configure hardware
 - **Type:** R&D Lab
 - **Date:** April 10, 2025
 - **Time:** 12:00 PM - 4:00 PM
 - **Duration:** 4h 0m
 - After the presentation we started configuring all the routers / PCs
 - **Total Hours for S1 - Week 6:** 9h 0m
-

S1 - Mid-Sem Break (April 14 - April 27, 2025)

- **Offsite:** Upskilling - Environment Setup
 - **Type:** Offsite
 - **Date:** April 14, 2025
 - **Time:** 12:00 PM - 4:00 PM
 - **Duration:** 4h 0m
 - Downloaded Virtual Box - To set up virtual machines
 - Downloaded the OS ISOs
 - Setup Ubuntu Server on 4 VMs
- **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** April 15, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Meeting Minutes 15-04-2025 \(Team\).docx](#)
 - The team reviewed their two-week upskilling plan, confirming the current project stage. This week, the focus is on Linux fundamentals and environment setup, followed by Linux routing and network testing in Week 2. A progress check-in is scheduled for the next weekly team meeting on April 22nd. Thomas has offered to provide informal support on Discord to address any skill discrepancies among members. The team will also familiarize themselves with D-ITG, the sole evaluation tool, by watching YouTube videos and reviewing the online manual.
- **Offsite:** Upskilling - Linux Fundamentals
 - **Type:** Offsite
 - **Date:** April 16, 2025
 - **Time:** 12:00 PM - 4:00 PM
 - **Duration:** 4h 0m
 - Linux Journey: <https://linuxjourney.com/>
 - LinuxCommand org: <https://linuxcommand.org/>
 - Command Line for Beginners: <https://www.freecodecamp.org/news/command-line-for-beginners/>
 - Explored these sites outlined in our upskilling document
- **Offsite:** Upskilling - Linux Fundamentals
 - **Type:** Offsite
 - **Date:** April 17, 2025
 - **Time:** 10:00 AM - 2:00 PM
 - **Duration:** 4h 0m
 - Linux Journey: <https://linuxjourney.com/>
 - LinuxCommand org: <https://linuxcommand.org/>
 - Command Line for Beginners: <https://www.freecodecamp.org/news/command-line-for-beginners/>
 - Explored these sites outlined in our upskilling document more as there was a lot of information

- **Offsite:** Upskilling - Networking Revision

- **Type:** Offsite
- **Date:** April 18, 2025
- **Time:** 10:00 AM - 4:00 PM
- **Duration:** 6h 0m
 - <https://www.freecodecamp.org/news/computer-networking-how-applications-talk-over-the-internet/>
 - <https://www.freecodecamp.org/news/linux-networking-commands-for-beginners/>
 - <https://www.freecodecamp.org/news/osi-model-networking-layers-explained-in-plain-english/>
 - [Upskilling - Networking Basics Revision.docx](#) - Took notes from the above sites

- **Offsite:** Upskilling - Linux as Router

- **Type:** Offsite
- **Date:** April 21, 2025
- **Time:** 10:00 AM - 4:00 PM
- **Duration:** 6h 0m
 - <https://fedoramagazine.org/use-fedora-server-create-router-gateway/>
 - <https://www.ibm.com/docs/en/linux-on-systems?topic=3-linux-as-router>
 - <https://www.computernetworkingnotes.com/linux-tutorials/how-to-configure-and-use-linux-as-a-router.html>
 - <https://www.baeldung.com/linux/server-router-configure>
 - <https://medium.com/@lfoster49203/setting-up-ubuntu-as-a-router-with-advanced-routing-features-4511abc5e1eb>
 - <https://www.tecmint.com/setup-linux-as-router/>
 - <https://wiki.archlinux.org/title/Router>
 - Used info from these sites to configure ubuntu as a router on the VMs. [IPv4 / IPv6]
 - Using NetPlan

- **Offsite:** Upskilling – D-ITG

- **Type:** Offsite
- **Date:** April 22, 2025
- **Time:** 12:00 AM - 2:00 PM
- **Duration:** 2h 0m
 - D-ITG: <https://github.com/jbucar/ditg>
 - [D-ITG-2.8.1-manual.pdf](#)
 - Gained somewhat of an understanding of D-ITG
 - Testing it on the 4 VMs

- **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** April 22, 2025
 - **Time:** 6:00 PM - 8:30 PM
 - **Duration:** 2h 0m
 - [Meeting Minutes 22-04-2025 \(Team\).docx](#)
 - This week, the team reviewed their progress on Week 1 upskilling, with each member discussing their achievements. While everyone successfully upskilled in Linux fundamentals, some struggled to find information about D-ITG and understand its manual. The team also discussed beginning test runs for the IPv4 team.
 - **Total Hours for S1 - Mid-Sem Break:** 30h 30m
-

S1 - Week 7 (April 28 - May 4, 2025)

- **Meeting:** Quick Meeting Catchup
 - **Type:** Meeting
 - **Date:** April 30, 2025
 - **Time:** 7:00 PM - 8:00 PM
 - **Duration:** 1h 0m
 - Quick catchup with the group before the 1st week back after mid-semester break
- **R&D Lab:** Configuration of Receiver
 - **Type:** R&D Lab
 - **Date:** May 1, 2025
 - **Time:** 9:00 AM - 1:00 PM
 - **Duration:** 4h 0m
 - Setting up IP addresses
 - Setup static routes between the PCs / Routers
 - Tested by pinging each other
- **Meeting:** Meeting Supervisor / Client
 - **Type:** Meeting
 - **Date:** May 1, 2025
 - **Time:** 1:00 PM - 2:00 PM
 - **Duration:** 1h 0m
 - [Meeting Minutes 01-05-2025 \(Client\).docx](#)
 - The Proposal Document needs to include a section on Technical Deliverables, specifically detailing the activities for Phase 1, such as providing test cases. For Functional Requirements, ensure you mention finding the optimal value first and synchronizing all computer clocks for accurate evaluation. When it comes to evaluation, plan to use a managed switch for IP address assignment, (which we decided we didn't need after the meeting) and note that setting up IPv4 and IPv6 concurrently isn't an issue. Start by finding the optimal value, perhaps with a 50,000 packet rate on TCP. For graphing, each graph should display all packet sizes, and you'll only need the receiver logs for the evaluation, which should be conducted one-way. Lastly, during your mid-project progress review, focus on discussing the changes made to the proposal.

- **R&D Lab:** Working on setup
 - **Type:** R&D Lab
 - **Date:** May 1, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - Discovered that we had some issues with the PCs freezing
 - Spent time troubleshooting
 - Discovered that it was a GPU driver issue
 - **Total Hours for S1 - Week 7:** 8h 0m
-

S1 - Week 8 (May 5 - May 11, 2025)

- **R&D Lab:** Work on trying to configure R1/R2
 - **Type:** R&D Lab
 - **Date:** May 8, 2025
 - **Time:** 9:00 AM - 2:00 PM
 - **Duration:** 7h 0m
 - Continued troubleshooting – Installed older drivers to see if it would fix the issue
 - Ended up having to set up SSH into the router PCs, to configure if the drivers break.
 - **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** May 8, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Meeting Minutes 2025-05-08 \(Team\).docx](#)
 - Key action items from the previous meeting, including the Ubuntu IPv4 Evaluation Runs and Sample Graphing Mock Data, were Extended. During the discussion, the IPv4 Team encountered issues with their initial setup using an NTP server and decided to troubleshoot the problem, look for an alternative (Chrony), and redo the evaluation. The IPv6 Team analysed mock data and was advised to look into tools like MATLAB or Python Scripts for graphing. New action items were assigned, including Setting Up Chrony NTP by Win Phyto, completing the Sample Graphing Mock Data by the IPv6 Team, and running the IPv4 TCP and UDP tests by Nathan, Win, and Thomas, all with an expected completion date of 15/05/2025.
 - **Total Hours for S1 - Week 8:** 7h 0m
-

S1 - Week 9 (May 12 - May 18, 2025)

- **Offsite:** Analysing the Data (IPv4 - 128)
 - **Type:** Offsite
 - **Date:** May 13, 2025
 - **Time:** 9:00 PM - May 14, 2025 2:00 AM
 - **Duration:** 5h 0m
 - Worked on creating a python script to extract the run data from the .dat files.
 - Used excel to make formulas to automatically average the data
 - **Offsite:** Analysing the Data
 - **Type:** Offsite
 - **Date:** May 14, 2025
 - **Time:** 4:00 PM - May 15, 2025 12:00 AM
 - **Duration:** 8h 0m
 - Worked on copying the formulas to all the other sheets for Delay, Jitter, Packetloss, and throughput.
 - **Offsite:** Analysing rest of the Data
 - **Type:** Offsite
 - **Date:** May 15, 2025
 - **Time:** 12:00 PM - 6:00 PM
 - **Duration:** 6h 0m
 - Finished what I hadn't finished the day before.
 - Added all the average data to one sheet to compare the data.
 - Noticed some issues with the data, so got the tests re-run
 - **Offsite:** Analysing re-done data
 - **Type:** Offsite
 - **Date:** May 16, 2025
 - **Time:** 4:00 PM - 11:00 PM
 - **Duration:** 7h 0m
 - With all the formulas already on the other sheets, I copied the sheets and just pasted the data. It then automatically did all the calculations.
 - **Total Hours for S1 - Week 9:** 26h 0m
-

S1 - Week 10 (May 19 - May 25, 2025)

- **Offsite:** Finished Analysing the data IPv4 Ubuntu data
 - **Type:** Offsite
 - **Date:** May 18, 2025
 - **Time:** 5:00 PM - 8:00 PM
 - **Duration:** 3h 0m
 - Found an issue where all the Run 5 data was broken for 256 packet size
 - Tests was re-run and inserted into the data
 - Turned all the data into graphs comparing the data. For IPv4 Ubuntu Server
 - **Offsite:** Analysed new data
 - **Type:** Offsite
 - **Date:** May 21, 2025
 - **Time:** 6:00 PM - 8:00 PM
 - **Duration:** 2h 0m
 - Analysed the new re-run data due to the massive outlier
 - **Meeting:** Client Meeting
 - **Type:** Meeting
 - **Date:** May 22, 2025
 - **Time:** 1:00 PM - 1:30 PM
 - **Duration:** 0h 30m
 - [Meeting Minutes 22-05-2025 \(Client\) .docx](#)
 - Daniel has provided feedback on the mid-term status report, advising that the milestone section should now be a simplified table reflecting only the current and upcoming tasks, rather than listing all phases. The Gantt chart needs to be moved earlier in the proposal, with a screenshot of current progress and labels referencing future sections, and critically, the entire appendix must be removed to keep the proposal at 7 pages. A new "Skills Adopted" column will replace "Time Spent" to highlight new resources, and a table will confirm the unchanged client-mentor meeting schedule. For IPv4 results, the report needs a clearer picture of completion and future plans like IPv6 testing, with a clear link between test objectives and the test case document, and it's important to note that jitter results shouldn't be linear. Both the status report and IPv4 results presentations should be limited to 6 slides, primarily featuring visuals and tables with minimal text, and a follow-up meeting with Raymond next week is suggested.
 - **Workshop:** Week 10 - Workshop
 - **Type:** Workshop
 - **Date:** May 22, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [Week 10 - Workshop Notes.docx](#)
 - **Total Hours for S1 - Week 10:** 7h 30m
-

S1 - Week 11 (May 26 - June 1, 2025)

- **Offsite:** Worked on Status Report
 - **Type:** Offsite
 - **Date:** May 26, 2025
 - **Time:** 2:00 PM - 6:00 PM
 - **Duration:** 4h 0m
 - Added to the Team Contribution for myself
 - Reviewed status report and worked on what was needing to be completed
- **Offsite:** Worked on Status Report
 - **Type:** Offsite
 - **Date:** May 27, 2025
 - **Time:** 2:00 PM - 6:00 PM
 - **Duration:** 4h 0m
 - Contributed to re-working the WBS
 - Contributed to Status Report
- **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** May 28, 2025
 - **Time:** 7:00 PM - 8:30 PM
 - **Duration:** 1h 30m
 - [Meeting Minutes 28-05-2025 \(Team\).docx](#)
 - D-ITG testing is progressing, with the team successfully receiving 10 simultaneous runs, though some finish earlier than others. For the status report, once the proposal errors are corrected, significant portions of the first section, including the executive summary, can be directly copied. The budget will also be refined by removing unnecessary equipment like an extra mouse, monitor, and gigabit adapters. The accompanying presentation will be concise, limited to 5 slides. Key proposal errors needing immediate attention include summarizing the Project Status, Risk, and Project Overview into a single paragraph, re-adding major milestones to the WBS, and correcting all dates within the schedule and milestones. Ultimately, the team needs to finalize the status report and comprehensively edit various proposal sections, including the Project Objective, Schedule, WBS, Gantt chart, Status, Milestones, Contribution, Recommendations, and Executive Summary.
- **Offsite:** Worked on Status report related docs
 - **Type:** Offsite
 - **Date:** May 27, 2025
 - **Time:** 9:00 AM - 4:00 PM
 - **Duration:** 7h 0m
 - Double checked the WBS to then start work on the new version of the Gantt Chart
 - Started the revised Gantt Chart based on the new WBS timeline
 - [09. Gantt Chart v2.mpp](#)
- **Total Hours for S1 - Week 11:** 16h 30m

S1 - Week 12 (June 2 - June 8, 2025)

- **R&D Lab:** Worked on preparing graphs for Client Meeting
 - **Type:** R&D Lab
 - **Date:** June 5, 2025
 - **Time:** 9:00 AM - 12:00 PM
 - **Duration:** 3h 0m
 - Prepared graphs from the IPv4 data that we had run to try and find the optimal value
 - **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** June 5, 2025
 - **Time:** 12:00 PM - 12:30 PM
 - **Duration:** 0h 30m
 - [Meeting Minutes 05-06-2025 \(Client\).docx](#)
 - Short meeting with Raymond. Going through the data we collected around the different amount for packets per second. Raymond said to use 120k packets per second as the graph looked correct. Talked about the confidence interval and he said that we can change it from the 95% interval in the brief and we need to calculate the confidence interval backwards from the data we have.
 - **R&D Lab:** More runs of evaluations
 - **Type:** R&D Lab
 - **Date:** June 5, 2025
 - **Time:** 12:30 PM - 6:00 PM
 - **Duration:** 5h 30m
 - With the optimal value of 120,000 packets per second now established, we re-ran our evaluations for TCP and UDP IPv4, and the results were sent to Raymond, who provided positive feedback. Following this, we moved on to IPv6 testing, which was also sent to Raymond for review and similarly received approval.
 - **Total Hours for S1 - Week 12:** 9h 0m
-

S1 - Week 13 (June 9 - June 15, 2025)

- **Offsite:** Finalizing status report
 - **Type:** Offsite
 - **Date:** June 14, 2025
 - **Time:** 11:00 PM – June 15 2025 3:00 AM
 - **Duration:** 4h 0m
 - Looked over all the changes making sure it was all done in the status report
 - Finalized the Gantt chart
 - [09. Gantt Chart v2.mpp](#)
- **Total Hours for S1 - Week 13:** 4h 0m

Total Hours End of Semester 1: 172 hours and 10 minutes

Mid-Year Break (July 17, 2025)

- **R&D Lab:** Assisting Installing Fedora
 - **Type:** R&D Lab
 - **Date:** July 17, 2025
 - **Time:** 2:00 PM - 6:00 PM
 - **Duration:** 4h 0m
 - Went in and assisted in installing fedora
 - Zaf and Larissa were the main people who were tasked with the set up.
-

Start of Semester 2:

S2 - Week 1 (July 21 - July 27, 2025)

- **R&D Lab:** Making sure Fedora is still working
 - **Type:** R&D Lab
 - **Date:** July 24, 2025
 - **Time:** 11:00 AM - 4:00 PM
 - **Duration:** 5h 0m
 - I tested that the first fedora install was still working. When checking it, there were some issues.
 - Nvidia driver kept freezing the output
 - Having to keep trying to get in to disable the driver from running manually
 - Having to re-install fedora as it had corrupted and broken.
 - Having to re-configure all the files so that the IPs are there in netplan.
 - When it kept freezing this worked to get past the freezing:
 - Adding “systemd.unit=multi-user.target” in the GRUB menu
 - **Total Hours for S2 - Week 1:** 5h 0m
-

S2 - Week 2 (July 28 - Aug 3, 2025)

- **Offsite:** Testing fedora config on VM
 - **Type:** Offsite
 - **Date:** July 29, 2025
 - **Time:** 3:00 PM - 5:00 PM
 - **Duration:** 2h 0m
 - Installed fedora on a VM
 - Tested using network manager instead of netplan since there were some issues with it not automatically applying.
 - Configured network manager and it all worked
- **R&D Lab:** Continuing to try and fix fedora systems
 - **Type:** R&D Lab
 - **Date:** July 31, 2025
 - **Time:** 9:00 AM - 4:00 PM
 - **Duration:** 7h 0m
 - Due to issues with installing netplan and getting it to keep the IP addresses, we have had to troubleshoot the issue.
 - This included reinstalling fedora a few times
- **Meeting:** Client Meeting
 - **Type:** Meeting
 - **Date:** July 31, 2025
 - **Time:** 1:30 PM - 2:00 PM
 - **Duration:** 0h 30m
 - reflective report - individual. REALLY LONG document, Daniel recommended to start now, section by section. reflection from logbook
 - poster, week 8 to discuss and plan the poster. after the mid-sem break. we should also finish testing by then A2 size. Daniel will bring examples
 - portfolio, MOST IMPORTANT. delete documents that aren't important.
 - client feedback. Raymond will give us written feedback 5% - there's a template
 - week 10
 - Mock presentation of the poster
 - real one in week 13
 - different set of people will be there
 - intro, requirement only words. More images. ensure it's easy to understand
 - No reply from Bobby - marks will be adjusted to 9.75
 - understandable if we do not meet full 300 hours due to the amount of members we have
- **Total Hours for S2 - Week 2:** 9h 30m

S2 - Week 3 (Aug 4 - Aug 10, 2025)

- **R&D Lab:** Practicing the kali install on VM
 - **Type:** R&D Lab
 - **Date:** August 7, 2025
 - **Time:** 11:00 AM - 4:00 PM
 - **Duration:** 5h 0m
 - Set up kali on 2x virtualbox VMs
 - Partial configuration of the 2x Kali VMs to be prepared for installation of Kali on the routers
 - Grabbed packages ready for kali and put them on a USB
 - Assisted with running fedora D-ITG runs
 - **Total Hours for S2 - Week 3:** 5h 0m
-

S2 - Week 4 (Aug 11 - Aug 17, 2025)

- **R&D Lab:** Finished Configuring Kali on VM
 - **Type:** R&D Lab
 - **Date:** August 14, 2025
 - **Time:** 9:00 AM - 1:30 PM
 - **Duration:** 4h 30m
 - Completed the rest of the configuration of the 2x kali VMs
- **Meeting:** Client Meeting
 - **Type:** Meeting
 - **Date:** August 14, 2025
 - **Time:** 1:30 PM - 2:00 PM
 - **Duration:** 0h 30m
 - Looked at Fedora Data and graphs
 - Client is fine with the data and happy to move onto Kali

- **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** August 14, 2025
 - **Time:** 2:00 PM - 2:30 PM
 - **Duration:** 0h 30m
 - 5 Graphs on A2 Poster
 - Happy with Fedora Graphs / Data
 - The data or graph has similar trends to Ubuntu
 - Need to find out about Confidence of Interval
 - Work on graphing Ubuntu and Fedora Data
 - May need to double check the Ubuntu Data
 - Ensure the size of the graph is big enough to accommodate all data
 - Starting Kali OS installation Next week (Charmi and Kylie)
 - Running Evaluation 5 runs (Charmi, Kylie, Nathan, Zaf, Larissa)
 - Graphing (Win Phyto, Nathan, Larissa)
 - Documentations + Poster (Thomas, Win Phyto, Zaf)
 - Start Individual Report for all members
 - Team meeting Next week Thursday
 - Client meeting (Possible) Week 6 Thursday
 - **R&D Lab:** Data Analysis
 - **Type:** R&D Lab
 - **Date:** August 14, 2025
 - **Time:** 2:30 PM - 4:00 PM
 - **Duration:** 1h 30m
 - Comparing the data from Ubuntu and Fedora
 - **Total Hours for S2 - Week 4:** 7h 0m
-

S2 - Week 5 (Aug 18 - Aug 24, 2025)

- **R&D Lab:** Initial Install of Kali
 - **Type:** R&D Lab
 - **Date:** August 21, 2025
 - **Time:** 9:00 AM - 4:00 PM
 - **Duration:** 7h 0m
 - Routers 1 / 2
 - Basic set up done Kali,
 - Kali updated,
 - Nvidia driver disabled,
 - Netplan installed (not configured),
 - Chrony installed (not configured)
 - This was done to make Charmi and Kylies life easier when they came to configuring the routers.
 - Charmi had a Family Emergency
 - I also worked on individual report
 - **Total Hours for S2 - Week 5:** 7h 0m
-

S2 - Week 6 (Aug 25 - Aug 31, 2025)

- **R&D Lab:** Running Kali Evaluations
 - **Type:** R&D Lab
 - **Date:** August 28, 2025
 - **Time:** 9:00 AM - 2:00 PM
 - **Duration:** 5h 0m
 - Ran the scripts to get the data needed to evaluate kali.
- **Workshop:** Workshop - Portfolio
 - **Type:** Workshop
 - **Date:** August 28, 2025
 - **Time:** 2:00 PM - 3:00 PM
 - **Duration:** 1h 0m
 - [S2 - Week 7 - Workshop Notes.docx](#)

- **Meeting:** Team Meeting
 - **Type:** Meeting
 - **Date:** August 28, 2025
 - **Time:** 3:00 PM - 4:00 PM
 - **Duration:** 1h 0m
 - Kylie, Charmi, and Nathan completed setting up and configuring Kali OS on both routers. Completed the evaluation 4 Runs, Decoding the logs and visualizing the data Planned to have Client meeting Next week for our graphs for Kali All team members should focus on doing the Individual Reflection Report at their own time. Week 8 (a week after the midterm break), we will start doing our final poster A3 Poster: Take everything so far and compress it down to fit the poster. Full of information with little text or words. Tasks Allegations will be discussed next Team meeting.
 - Getting everyone involved
 - Refrain from using AI for any summarizing Team decided on finalizing everything by at least week 10 to be fully prepared for mock presentation (Week 11 or 12) Look into GitHub for version control and for our Portfolio folders Also, on setting up Hyperlinks for Documents.
- **Offsite:** Graphing Data from Kali
 - **Type:** Offsite
 - **Date:** August 28, 2025
 - **Time:** 11:00 PM - 2:00 PM
 - **Duration:** 3h 0m
 - After doing runs 1 and 2 there were certainly some discrepancies with the data
 - When doing runs 3 - 5 it was more the same kind of pattern which we normally saw throughout the other runs on the other OS's
- **Total Hours for S2 - Week 6:** 10h 0m

S2 - Week 7 (Sep 1 - Sep 7, 2025)

- **Offsite:** Graphing Data
 - **Type:** Offsite
 - **Date:** September 3, 2025
 - **Time:** 8:00 PM - 10:00 PM
 - **Duration:** 2h 0m
 - Graphed the new kali runs (6 - 8)
 - Compared data of runs from Fedora and Kali
- **R&D Lab:** Graphing the compare data
 - **Type:** R&D Lab
 - **Date:** September 4, 2025
 - **Time:** 9:00 AM - 1:30 PM
 - **Duration:** 4h 30m
 - Creating all the graphing off the combined data.

- **Meeting:** Client Meeting
 - **Type:** Meeting
 - **Date:** September 4, 2025
 - **Time:** 1:30 PM - 2:00 PM
 - **Duration:** 0h 30m
 - Condense graphs into 1 data, show differences (graph),
 - Get questions like “why is this not on poster” (out of scope),
 - With graphs, stick with latest one.
 - Raymond says graphs look good, consistent.
 - Example poster:
 - Methodology diagram and explain, gives more space for graphs,
 - Look at marking criteria,
 - Challenges, main thing u dealt w whole year, technical/non tech,
 - Lessons learned, what we learned as a group,
 - QR code for design, look at other data
 - Portfolio, link between all documents, turn into website.
 - Put everything in 1 folder,
- **Workshop:** Workshop - Critical reflection
 - **Type:** Workshop
 - **Date:** September 4, 2025
 - **Time:** 2:00 PM - 4:00 PM
 - **Duration:** 2h 0m
 - [S2 - Week 7 - Workshop Notes.docx](#)
- **Total Hours for S2 - Week 7:** 7h 0m

S2 - Mid-Sem Break (Sep 14 - Sep 20, 2025)

- **Offsite: Working on individual reflection report**
 - Type: Offsite
 - Date: September 18, 2025
 - Time: 6:00 PM - 11:00 PM
 - Duration: 5h 0m
- **Offsite: Generating graphs using python**
 - Type: Offsite
 - Date: September 19, 2025
 - Time: 6:00 PM - 9:00 PM
 - Duration: 3h 0m
 - Generating graphs using python

Total Hours for S2 - Mid-Sem Break: 8h 0m

S2 - Week 8 (Sep 21 - Sep 27, 2025)

- **R&D Lab: Double checked the data and graphing / Installing Ubuntu Again**
 - Type: R&D Lab
 - Date: September 25, 2025
 - Time: 9:00 AM - 2:00 PM
 - Duration: 5h 0m
 - When double checking the data I discovered that there were some issues with the ubuntu data.
 - When checking it there were some spikes.
 - We discovered that the data wasn't the up to date data and it was deleted ages ago
 - We plan to install ubuntu again and configure it. So that we can get some new data.
 - Installed and configured R1 and R2
 - Started running tests again
 - Run 5 tests
- **Workshop: Poster**
 - Type: Workshop
 - Date: September 25, 2025
 - Time: 2:00 PM - 4:00 PM
 - Duration: 2h 0m
 - [S2 - Week 8 - Workshop Notes.docx](#)
- **Offsite: Uploading and decoding new ubuntu runs**
 - Type: Offsite
 - Date: September 27, 2025
 - Time: 11:00 AM - 2:00 PM
 - Duration: 3h 0m
 - Uploaded all the new ubuntu logs and decoded them to onedrive

Total Hours for S2 - Week 8: 10h 0m

S2 - Week 9 (Sep 28 - Oct 4, 2025)

- **R&D Lab: Comparing data from the ubuntu runs**
 - Type: R&D Lab
 - Date: October 2, 2025
 - Time: 9:00 AM - 1:30 PM
 - Duration: 4h 30m
 - Compared the data from the new ubuntu logs to the other OS logs.

- **Meeting: Client Meeting**

- Type: Meeting
- Date: October 2, 2025
- Time: 1:30 PM - 2:00 PM
- Duration: 0h 30m
 - The meeting focused on refining the project's data analysis and presentation. A clear Out of Scope Section is necessary in the documentation to define the project's boundaries for all readers. For the Mock-up Poster Draft, the final visual presentation should include the Project Methodology Diagram and a maximum of two graphs illustrating Throughput and Delay. While the team should be ready to present all supplementary work (graphs, Excel sheets, scripts) and understand the basic characteristics of networking protocols like TCP and UDP to explain the data trends, detailed reasoning behind the data's results is not required. The overall methodology is strictly limited to the documented tasks and client requests, and anything else is considered out of scope or not supported by the project data, which is to be included with the latest raw data and Excel sheets in the portfolio.

- **Workshop: Project Handover**

- Type: Workshop
- Date: October 2, 2025
- Time: 2:00 PM - 4:00 PM
- Duration: 2h 0m
 - [S2 - Week 9 - Workshop Notes.docx](#)

Total Hours for S2 - Week 9: 7h 0m

S2 - Week 10 (Oct 5 - Oct 11, 2025)

- **Offsite: Reflection Report**

- Type: Offsite
- Date: October 5, 2025
- Time: 10:00 PM - 3:00 AM
- Duration: 5h 0m
 - Worked on Reflection Report

- **Offsite: Reflection Report**

- Type: Offsite
- Date: October 6, 2025
- Time: 2:00 PM - 6:00 PM
- Duration: 4h 0m
 - Worked on Reflection Report

- **R&D Lab: Reflection Report**

- Type: R&D Lab
- Date: October 9, 2025
- Time: 9:00 AM - 4:00 PM
- Duration: 7h 0m
 - Worked on Reflection Report

Total Hours for S2 - Week 10: 16h 0m

S2 - Week 11 (Oct 12 - Oct 18, 2025)

- **Offsite: Worked on each dataset for OS and making graphs**

- Type: Offsite
- Date: October 13, 2025
- Time: 11:00 AM - 4:00 PM
- Duration: 5h 0m
 - Based on the data sets created the graphs and verified the data.

- **Meeting: Online Meeting**

- Type: Meeting
- Date: October 15, 2025
- Time: 5:00 PM - 5:30 PM
- Duration: 0h 30m
 - Online team meeting covered presentation preparation, including a review of the poster and the final allocation of speaking sections: Thomas will cover the introduction, team members, background, and rationale; Nathan has the objectives; Win is responsible for artefacts and graphs; Charmi will present on project impact and lessons learned; Kylie takes goals and key challenges; Larissa is allocated quality assurance; and Zaf will cover the methodology. A practice run is scheduled for the next day at 11:00 AM, and the immediate to-do list focuses on final documentation cleanup, which includes reviewing the Issue Logs, ensuring Meeting Minutes are up-to-date and linked, conducting an overall review of all project documentation for completeness, continuing work on individual reflective reports, and ultimately completing the Portfolio.

- **Offsite: Portfolio Website Design**

- Type: Offsite
- Date: October 15, 2025
- Time: 8:00 PM - 10:00 PM
- Duration: 2h 0m
 - Assisted in designing the website on the portfolio hosted on GitHub

- **Meeting: Client Meeting (Poster Presentation Practice)**

- Type: Meeting
- Date: October 16, 2025
- Time: 2:30 PM - 3:00 PM
- Duration: 0h 30m
 - Goals: Must be moved above Objectives and simplified to reflect only the core goal of the project.
 - Background: Must be very short.
 - Rationale: Must be limited to one sentence.
 - Objectives: The full description must be retained: "The objectives of the project were evaluating 3 Linux OS's (Ubuntu, Fedora, and Kali) across IPv4 and IPv6 using TCP and UDP protocols, measuring throughput, delay, jitter, and packet loss across packet sizes from 128 to 1536 with 10 runs for each evaluation."
 - Technical Parameters to Display: Clearly show IPv4 / IPv6, TCP / UDP, throughput, delay, jitter, packet loss, packet sizes 128 to 1536, and 10 runs per evaluation (potentially in a list or code-like format for clarity).
 - Project Impact: This section is to be removed and replaced with a slide covering Functional and Non-Functional Requirements.
 - Methodology: Keep the discussion minimal.
 - Artifacts: Remove all text; display only images (graphs) with only a few key points.
 - QA (Quality Assurance): Discuss only the high-level aspects.
 - Key Challenges: Discuss only the key points and keep it brief.
 - Lessons Learnt: Replace the focus on communication with a discussion on how the methodology received client approval or other client validation, keeping it brief.
 - Miscellaneous: The presentation must include only one QR code.

- **Meeting: Team Meeting Debrief**

- Type: Meeting
- Date: October 16, 2025
- Time: 3:00 PM - 4:00 PM
- Duration: 1h 0m
 - [Meeting Minutes 2025-10-16 \(Team\).docx](#)
 - Key changes for the poster included keeping the background short, having a one-sentence rationale, placing goals before objectives, replacing "Project Impact" with Functional/Non-Functional Requirements, and using the term EVALUATION instead of "TESTING". Team members were advised to shorten their presentation script and begin the Individual Reflective Report, which is due on October 28, 2025. Several action items were assigned with an expected completion date of October 31, 2025, including reformatting the poster layout by Larissa and shortening individual sections by All team Members. The next meeting is scheduled for October 28, 2025, at 5 PM to discuss finalising the team portfolio.

Total Hours for S2 - Week 11: 9h 0m

S2 - Week 12 (Oct 20 - Oct 24, 2025)

- **R&D Lab: Reflection Report / Update Logbook in Word**

- Type: R&D Lab
- Date: October 23, 2025
- Time: 9:00 AM - 4:00 PM
- Duration: 7h 0m
 - Worked on Reflection Report
 - Updated log book into word document

Total Hours for S2 - Week 12: 7h 0m

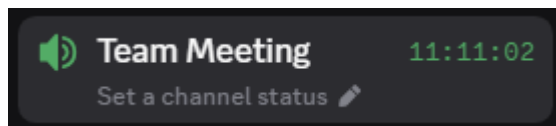
S2 - Week 13 (Oct 27 - Oct 31, 2025)

- **Meeting: Team Meeting - Checking Portfolio Documents**

- Type: Meeting
- Date: October 27, 2025
- Time: 7:00 PM - 12:00 AM
- Duration: 5h 0m
 - [Meeting Minutes 2025-10-27 \(Team\).docx](#)
 - The main focus of the meeting was to debrief and discuss any remaining tasks and issues. A major problem was brought to attention: a teammate had accidentally overwritten all client-mentor meeting entries. Nathan, Larissa, and Win spent time restoring these entries, ensuring all team meeting minutes were correctly uploaded and updated, and instituting backups to prevent future loss.
 - A major issue was addressed and resolved during the discussion: a teammate had overwritten all client-mentor meeting entries, requiring Nathan, Larissa, and Win to spend time restoring the entries, uploading all necessary minutes, and implementing backups. Remaining action items, many now due on 31/10/2025, include the Individual Reflective Report for all members and final sections of the Poster, while the final meeting preparation tasks (like creating a list of final changes) were completed. The next meeting is set for 28/10/2025 to finalize everything for the final portfolio submission.

- **Meeting: Team Meeting - Worked on missing portfolio documents**

- Type: Meeting
- Date: October 28, 2025
- Time: 12:00 PM – 11:00 PM
- Duration: 11h 0m
 - Going over and adding the last of the missing files.



Total Hours for S2 - Week 13: 16h 0m

Total Hours for Semester 2 – 127 hours 30 minutes

Summary:

Total Hours End of Semester 1: 172 hours and 10 minutes

Total Hours End of Semester 2: 129 hours and 30 minutes

Total Hours for the Year - 301 hours and 40 minutes.